Supplementary Material
Beautified Is Not Forged: Measuring and Separating Beautification from Face Forgery

S1 Pre-registration and what was read when

Every bar quoted in the paper was written into a dated file before the corresponding model was trained or the corresponding test source was read. Celeb-DF-B and the DFD filter set were read once per model, after the checkpoint had been chosen on validation data. The Celeb-DF-v2 filter set was used to diagnose which filters cause the damage (Sec. 4.2 of the paper), so it is a development source for every design choice made after that point; the DFD filter set was built afterwards from a second filter library. No per-filter result of the DFD set was inspected. The checkpoint of every model is chosen on validation data only: FF++ validation macro-F1 (three-way, FF++ only), FF++ validation AUC (binary), or the mean of FF++ validation macro-F1 and balanced accuracy on held-out originals and renders of the two training services (three-way, full data). An earlier selection rule (F1 of the filter class on an imbalanced validation set) picked the epoch with the lowest specificity on untouched originals; it was replaced before the models of this paper were trained, and the earlier model appears only in Table S8.

S2 Each baseline under its own preprocessing

Table S1: Reported and reproduced. Published numbers are from DeepfakeBench (frame level, eight CNNs) and from the Effort and Forensics Adapter papers. "DFB crop": our run of the released weights on DeepfakeBench crops (dlib frontal detector, 81 landmarks, five-point similarity alignment, scale 1.3, 256 px, 32 frames per video by linear spacing), built with the benchmark's own alignment function. "Our crop": landmark box with a 35% margin, aspect kept. Videos common to both frame sets (516 for Celeb-DF-v2, 332 for DFD). Our DFD set is a 332-video subset, so no published DFD number is comparable and none is listed as a target.

	Celeb-DF-v2 frame AUC		Celeb-DF-v2 video AUC		DFD frame AUC	
Detector	published	ours, DFB crop / our crop	published	ours, DFB crop / our crop	published	ours, DFB crop / our crop
Xception	0.737	0.744 / 0.718	–	0.821 / 0.772	–	0.845 / 0.854
EfficientNet-B4	0.749	0.754 / 0.750	–	0.813 / 0.807	–	0.818 / 0.810
F3Net	0.735	0.732 / 0.708	–	0.793 / 0.750	–	0.781 / 0.781
SPSL	0.765	0.732 / 0.703	–	0.803 / 0.752	–	0.812 / 0.829
UCF	0.753	0.775 / 0.744	–	0.841 / 0.788	–	0.818 / 0.822
RECCE	0.732	0.744 / 0.688	–	0.827 / 0.731	–	0.839 / 0.833
CORE	0.743	0.744 / 0.715	–	0.814 / 0.764	–	0.830 / 0.827
SRM	0.755	0.763 / 0.717	–	0.844 / 0.778	–	0.837 / 0.830
Effort [1]	0.901	0.879 / 0.881	0.956	0.927 / 0.930	0.923	0.933 / 0.939
Forensics Adapter [2]	0.900	0.899 / 0.887	0.957	0.956 / 0.945	0.933	0.940 / 0.940
GenD [3]	–	0.900 / 0.891	0.960	0.960 / 0.954	–	0.932 / 0.934
DFD-HR [4]	0.910	0.886 / 0.879	0.960	0.951 / 0.943	0.953	0.942 / 0.943

Forensics Adapter reproduces its published Celeb-DF-v2 numbers on its own crop. Effort does not; an independent re-run of its released weights reports 0.932 at video level and a retraining 0.871, against 0.956 in the paper, and our 0.927 is in line with the re-run. The training code released with both detectors keeps the checkpoint with the best test-set AUC; the papers do not state a selection rule. All comparisons in the paper pair our models on our crop with each baseline on its own crop.

S3 Where the filter is applied

Table S2: The same videos, frames and per-video filter assignment with the filter applied to the face crop or to the full frame before the crop. A filter with a vignette is centred on the face in the first case. The paper reports the full-frame version.

	Celeb-DF-v2, 11 filters			DFD, 16 filters		
Detector	AUC	AUC_f crop	AUC_f full frame	AUC	AUC_f crop	AUC_f full frame
Xception	0.718	0.673	0.693	0.854	0.823	0.834
EfficientNet-B4	0.750	0.698	0.725	0.810	0.777	0.781
F3Net	0.708	0.663	0.688	0.781	0.769	0.774
SPSL	0.703	0.664	0.683	0.829	0.834	0.837
UCF	0.744	0.669	0.697	0.822	0.793	0.804
RECCE	0.688	0.636	0.661	0.833	0.795	0.808
CORE	0.715	0.669	0.693	0.827	0.813	0.821
SRM	0.717	0.696	0.712	0.830	0.811	0.822
SBI [5]	0.817	0.729	0.776	0.829	0.803	0.810
UnivFD	0.662	0.610	0.622	0.840	0.793	0.801
NPR	0.522	0.502	0.497	0.473	0.541	0.516
Effort [1]	0.881	0.788	0.813	0.939	0.904	0.905
Forensics Adapter [2]	0.887	0.790	0.814	0.940	0.907	0.924
GenD [3]	0.891	0.818	0.856	0.934	0.913	0.925
DFD-HR [4]	0.879	0.781	0.824	0.943	0.921	0.928
OURS	0.891	0.849	0.873	0.942	0.927	0.931
OURS, FF++ only	0.898	0.797	0.851	0.943	0.925	0.932
binary control	0.914	0.863	0.883	0.961	0.949	0.952

S4 Exposure arms on every source

Table S3: Which class receives beautified training views (P0 none, P1 genuine only, P2 both), FF++ only, forged head only; values per seed. Effort scheme: the SVD-residual tuning of the released Effort code in place of LoRA. Filters on the full frame; ΔMISS at the 5%-FPR threshold of untreated genuine frames.

		Celeb-DF-B			Celeb-DF-v2 filters		DFD filters	
Tuning	Arm	AUC	AUC_b	$\Delta\text{MISS}_{\text{beauty}}$	AUC_f	ΔMISS	AUC_f	ΔMISS
LoRA	P0	0.927 / 0.937 / 0.944	0.860 / 0.883 / 0.883	+17.0 / +14.2 / +10.8	0.852 / 0.858 / 0.855	+7.1 / +4.4 / +8.3	0.924 / 0.928 / 0.932	+9.3 / +7.2 / +8.7
LoRA	P1	0.928 / 0.931 / 0.939	0.844 / 0.851 / 0.864	+21.5 / +21.8 / +20.4	0.798 / 0.820 / 0.838	+16.6 / +17.7 / +16.3	0.905 / 0.911 / 0.913	+11.2 / +9.3 / +9.2
LoRA	P2	0.929 / 0.933 / 0.935	0.879 / 0.893 / 0.877	+9.8 / +10.3 / +12.8	0.848 / 0.850 / 0.856	+0.3 / +3.0 / -0.9	0.929 / 0.929 / 0.927	+2.5 / +1.8 / -0.5
Effort scheme	P0	0.943 / 0.948	0.882 / 0.886	+10.1 / +10.7	0.876 / 0.886	+2.3 / -1.3	0.942 / 0.947	+8.7 / +7.4
Effort scheme	P2	0.938 / 0.943	0.889 / 0.890	+5.5 / +5.5	0.872 / 0.879	-7.6 / -8.4	0.948 / 0.953	+3.0 / +2.4

Paired video-cluster intervals for P2 minus P0 on the two filter sources: the miss rise falls by 5.0 to 9.9 points in nine of ten seed-by-source cells (intervals below zero; the tenth is -1.4 $[-3.3, 0.6]$), the false-alarm rise grows by 2.0 to 4.6 points in the same nine, and the AUC under the filters changes by -0.008 to $+0.006$ (intervals contain zero except for the Effort scheme on DFD, $+0.006$). For P1 minus P0 on Celeb-DF-v2 the miss rise grows by 13.3 and 8.0 points and the AUC under the filters falls by 0.038 and 0.017.

S5 Linear erasure of the beautification subspace

The two leading directions hold 79% of the shift energy of training-time beautifications. Removing them costs 0.05 clean AUC and does not lower the false alarms under the held-out filters, so the directions that beautification moves along carry forgery evidence.

Table S4: FF++-only CLIP model, FF++ validation frames under the eleven held-out filters. The feature shifts caused by training-time beautifications are decomposed by SVD on training frames, the first k directions are projected out, and a logistic forged head is refitted on training features. FA and MISS at the 5%-FPR threshold of clean genuine frames.

Forged head	AUC clean	AUC filtered	FA filtered	MISS filtered
trained head	0.987	0.953	21.5	5.9
refit, 0 directions erased	0.985	0.916	15.8	16.6
refit, 2 directions erased	0.933	0.907	18.5	15.1
refit, 8 directions erased	0.932	0.844	18.3	27.6
refit, 32 directions erased	0.919	0.804	19.2	32.1

Table S5: Every method arm tried under a pre-registered bar. FF++ only unless noted; each block starts with its reference. Bars for the FF++-only arms: AUC_b on Celeb-DF-B ≥ 0.890 and $\Delta\text{MISS}_{\text{beauty}} \leq 7.5$, untreated Celeb-DF-v2 AUC within 0.01 of the reference. None is met.

	Celeb-DF-B				Celeb-DF-v2	DFD filters
Arm	AUC	AUC_b	$\Delta\text{MISS}_{\text{beauty}}$	FA_b	AUC untreated	AUC_f
random colour grading (reference)	0.938	0.872	+7.5	7.8	0.898	0.932
no grading	0.929	0.849	+14.1	7.2	0.880	0.917
operator, random edits	0.932	0.876	+7.9	7.7	0.873	0.932
operator, worst of 4 draws	0.928	0.867	+12.9	5.6	0.860	0.935
operator, 1-step search	0.930	0.864	+8.2	8.3	0.876	0.935
operator, search within measured edits	0.928	0.869	+15.1	5.2	0.874	0.934
operator, consistency objective	0.924	0.856	+13.0	5.7	0.849	0.922
operator, 3-step search	0.922	0.859	+8.9	7.6	0.849	0.937
decision pooled from the evidence map	0.937	0.877	+14.9	5.0	0.889	0.924
same, no beautified views	0.942	0.891	+14.7	4.4	0.899	0.932
full data, binary (reference)	0.949	0.896	+7.0	7.4	0.914	0.952
full data, binary, operator search	0.942	0.892	+9.7	6.0	0.911	0.945
released Effort, mapped (reference)	0.909	0.808	+11.2	4.7	0.886	0.904
released Effort + 2 epochs, no beautified views	0.903	0.805	+5.8	9.4	0.866	0.918
released Effort + 2 epochs, both classes	0.895	0.803	+2.7	10.3	0.861	0.916

S6 Methods that did not meet their bars

The operator. A differentiable edit $B(x; u)$ on a face crop in $[0, 1]$ with every component bounded by a sigmoid: per-channel gamma, gain and lift; a 3×3 colour matrix close to the identity; saturation from monochrome to 1.7; contrast about the mean; a 4×4 luminance field and a per-channel tint field, upsampled bilinearly; a blend toward a Gaussian-smoothed copy; additive grain. Its parameters were fitted by gradient descent to 1,000 pairs each of four measured families (Table S6); the search of the "measured edits" arms starts from a fitted edit and is projected onto the 0.5th–99.5th percentile range of the fitted parameters.

Table S6: The operator fitted to measured edits (training split only). The residual of the commercial renders is geometry, which a photometric operator cannot express.

Measured edit	n	PSNR before	PSNR after fit	10th pct
Tencent retouching	1000	21.4	28.6	24.8
Megvii retouching	1000	29.1	30.6	28.0
training colour filters	1000	21.0	40.1	32.0
random grading	1000	24.2	48.4	39.0

The search works; the training does not transfer. On 400 FF++ validation frames the reference model keeps 96.0% of genuine faces genuine without an edit, 71.0% under a random draw of the operator, 41.0% under the worst of four draws, 8.5% under a one-step signed-gradient search and 0.0% under ten steps; forged faces stay forged in 93.5, 89.5, 83.0, 74.0 and 20.0% of cases. After training on the one-step worst case the genuine accuracy under that search is 71.5% (25.0% under ten steps), and after the three-step variant 76.5% (41.5%). The gain does not carry over to the app presets or to filters from a library the operator was not fitted to, and the forged accuracy under random edits falls from 89.5 to 79.5%: with a label loss, edited views of genuine faces dominate the signal and the model learns that a heavily edited face is less likely

forged.

A learned inverse. The exact algebraic inverse of the invertible part of the operator restores an edited frame to 53–58 dB when the true parameters are given. A ResNet-18 estimator trained to predict them from the edited frame alone reaches 26.7 dB on random edits (25.9 before) and 22.7 dB on the held-out filters (21.0 before), and moves untouched frames to 34.4 dB. In front of the detector it changes the false alarms under the held-out filters from 21.5 to 23.6%. A single image does not determine its colour transform.

Edit sensitivity as evidence. The response of the forged logit to six fixed probe edits is larger for genuine than for forged frames on average, but inside the deciles of the clean logit where both classes occur the two respond alike (1.15 vs 0.96; 0.48 vs 0.10), so the asymmetry is a ceiling effect; adding the responses to the logit leaves the false alarms under filters at 16–22%.

A decision pooled from the evidence map. With a zero map target on every beautified item and the forged logit taken as the mean of the top 5% map cells, the map stays dark on filtered genuine faces (mean logit -10.2 against $+1.9$ for the edit map) and the false alarms under filters fall from 21.5 to 14.7%, but the beautification-specific miss rise on Celeb-DF-B grows from 7.5 to 14.9 points.

Fine-tuning a released checkpoint. The released Effort weights were loaded into our pipeline (the mapped model reproduces the released scorer: Celeb-DF-B AUC 0.909, AUC_b 0.808) and fine-tuned for two epochs. With both-class exposure the miss rise falls from 11.2 to 2.7 points, the false alarms on beautified genuine faces rise from 4.7 to 10.3%, AUC_b stays at 0.803 and the untreated AUC falls by 0.014; without beautified views the same fine-tuning gives 5.8 points and 9.4%. The operating point moves; separability does not.

S7 Commercial retouching: all variants

Table S7: All trained variants on the unseen service (2,210 originals, 26,519 renders). Columns as in Table V of the paper.

	originals	retouched (pooled)				paired rank at level 90	smoothing-90
Model	real/fake/filter	→filter	→fake	balanced	paired rank	eye / white / lift / smooth	→fake
FF++ c23 training only (zero-shot)							
three-way, RepViT-M0.9	0.8/2.5/96.7	92.6	6.4	47.9	48.4	73 / 55 / 52 / 15	17.7
three-way, observed fakes (SUP)	36.8/5.7/57.5	66.5	9.1	54.5	79.8	97 / 89 / 79 / 59	27.3
three-way, hybrid fakes (HYB)	33.8/16.7/49.5	54.6	22.1	52.6	69.2	92 / 83 / 65 / 44	50.2
HYB, faithful blend	28.2/22.1/49.6	52.8	28.0	51.6	65.7	92 / 77 / 62 / 35	58.4
HYB + symmetric degradation	31.7/26.3/42.0	45.3	33.6	51.7	64.3	89 / 76 / 63 / 35	63.1
three-way, part edits + evidence head	16.7/55.9/27.4	28.6	59.4	50.6	60.9	89 / 78 / 60 / 20	85.8
Vendor-trained variants (all single Alibaba reads)							
RU1 renders only (EffNet-B4, s1)	93.4/0.8/5.8	38.1	0.6	66.1	92.8	99 / 85 / 95 / 100	0.1
RU1 renders only (RepViT)	92.8/1.4/5.9	36.1	1.1	65.1	92.6	96 / 89 / 93 / 100	0.3
RU2 + components (EffNet-B4, s1)	76.8/0.5/22.8	68.2	0.3	72.7	94.2	99 / 90 / 94 / 100	0.4
RU2 + components (EffNet-B4, s2)	66.6/0.3/33.2	74.9	0.2	70.8	94.2	99 / 92 / 93 / 99	0.4
RU2 + components (RepViT)	43.4/0.9/55.7	83.7	0.5	64.0	95.1	99 / 91 / 96 / 99	1.1
RU3 cleaned components (EffNet-B4, s1)	76.7/0.4/22.9	68.8	0.3	72.9	94.9	99 / 92 / 94 / 99	0.5
RU3 + evidence head (EffNet-B4, s1)	83.2/0.6/16.2	62.2	0.5	73.0	94.1	99 / 91 / 93 / 99	0.8
RU4 + part edits (lite) (EffNet-B4, s1)	81.0/0.4/18.6	63.8	0.3	72.6	93.8	99 / 91 / 93 / 99	0.4
RU3 + evidence head, CLIP (CLIP-L/14, s1)	77.4/0.4/22.2	71.4	0.2	74.6	93.8	98 / 90 / 94 / 100	0.2
RU3 + evidence head, CLIP (CLIP-L/14, s2)	73.0/0.3/26.7	73.6	0.2	73.4	95.1	99 / 94 / 96 / 100	0.2
RU4 + part edits, CLIP (filter-F1 selection) (CLIP-L/14, s1)	57.9/0.3/41.9	82.3	0.1	70.2	94.8	98 / 94 / 95 / 100	0.0
RU4 + part edits, CLIP (CLIP-L/14, s1)	89.5/0.6/9.8	59.2	0.5	74.7	95.5	98 / 94 / 96 / 100	0.1
RU4 + part edits, CLIP (CLIP-L/14, s2)	83.3/0.4/16.2	67.5	0.2	75.6	93.0	98 / 88 / 95 / 100	0.2
RU5 without colour grading (CLIP-L/14, s1)	82.1/0.6/17.3	65.7	0.3	74.2	92.8	97 / 89 / 93 / 99	0.4
RU5 without vendor edit on forgeries (CLIP-L/14, s1)	83.8/0.4/15.8	65.6	0.2	74.9	94.7	98 / 94 / 94 / 100	0.1
RU5 data, three-way softmax (CLIP-L/14, s1)	81.5/0.7/17.8	66.4	0.5	74.3	94.1	98 / 94 / 93 / 99	0.5
RU5, CLIP (main) (CLIP-L/14, s1)	78.4/0.8/20.8	67.7	0.6	73.5	92.9	98 / 92 / 94 / 99	0.8
RU5, CLIP (main) (CLIP-L/14, s2)	79.8/0.5/19.6	70.5	0.4	75.4	94.3	98 / 92 / 95 / 99	0.6

S8 Explanations: all variants

The donor-transplant test set uses donors from test videos only. An earlier version of that set drew donors from training videos (1,023 donor-frame and part pairs occurred in training); it was rebuilt, and the numbers of the main model moved by less than 0.2 points.

Table S8: Restoration test on held-out part-level forgeries for every variant. Columns as in Table VI of the paper.

	donor transplant				SD-1.5 inpaint				SDXL inpaint (unseen editor)			
Map	IoU_7	flip	ceil.	compl.	IoU_7	flip	ceil.	compl.	IoU_7	flip	ceil.	compl.
centre prior / random region (last col.)	0.310	38.6	–	17.0	0.249	40.9	–	17.9	0.255	40.2	–	17.7
no head, Grad-CAM	0.367	50.9	85.8	15.2	0.259	45.4	85.2	25.0	0.269	48.0	85.9	23.6
no head, Grad-CAM++	0.414	51.7	85.8	7.2	0.327	47.8	85.2	16.7	0.330	48.8	85.9	19.1
no head, Score-CAM	0.408	56.5	85.8	5.9	0.334	52.4	85.2	15.4	0.341	52.4	85.9	17.1
head, part edits, FF++ only	0.723	84.0	84.7	0.4	0.752	82.6	83.2	0.7	0.746	82.9	83.8	0.7
+ symmetric degradation	0.723	86.3	87.1	0.4	0.719	82.5	83.7	2.0	0.728	83.7	84.4	1.5
+ restoration-consistency loss	0.759	89.4	90.0	0.4	0.614	75.2	79.8	4.4	0.684	82.5	84.6	2.6
+ consistency + symmetric degradation	0.722	88.4	89.3	0.7	0.723	85.2	86.1	2.4	0.732	86.1	86.9	1.8
head, vendor data, no part edits (EffNet-B4)	0.418	61.4	82.4	12.0	0.280	39.5	63.5	23.4	0.315	46.2	72.3	22.5
head, vendor data, no part edits (CLIP)	0.479	78.8	89.4	2.5	0.376	53.9	69.1	14.4	0.445	68.9	78.3	8.4
OURS-lite (EffNet-B4)	0.807	87.9	88.0	0.0	0.820	87.4	87.6	0.4	0.812	86.9	87.2	0.4
RU4, CLIP (filter-F1 selected, superseded)	0.832	91.7	91.9	0.1	0.845	91.5	91.6	0.2	0.827	90.9	91.2	0.5
OURS (CLIP ViT-L/14 + LoRA)	0.834	94.0	94.1	0.1	0.841	93.6	93.7	0.4	0.831	93.5	93.6	0.3

S9 Sources that were examined and excluded

AR beauty filters. On paired Instagram AR-filter images, a model trained with such pairs called 94.8% of the filtered images *filter* with the face hull greyed out: the capture pipeline, not the face, carries the label. No claim is made on AR filters. **WildDeepfake.** The 800 test frames available to us come from six sequences, which gives intervals of 0.3–0.5 AUC; not reported. **DFDC, DFDCP, DF40.** Not available to us in a form with genuine and forged frames from the same pipeline.

References

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